Finetuning ProGen2 on Green Fluorescent Protein

Assignment One: Top-5 Candidate Report

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Top-5 Candidates

Rank	Design	Mean pLDDT	pTM	Substitutions
1	Design_022	92.52	0.92	A36V, T37A, S64A
2	Design_027	92.35	0.92	A36T, V54M
3	Design_018	92.30	0.92	A36V, Y38H
4	Design_092	92.28	0.92	A36V
5	Design_061	92.26	0.92	A36D, Y38N, T62A

Metrics Reasoning:

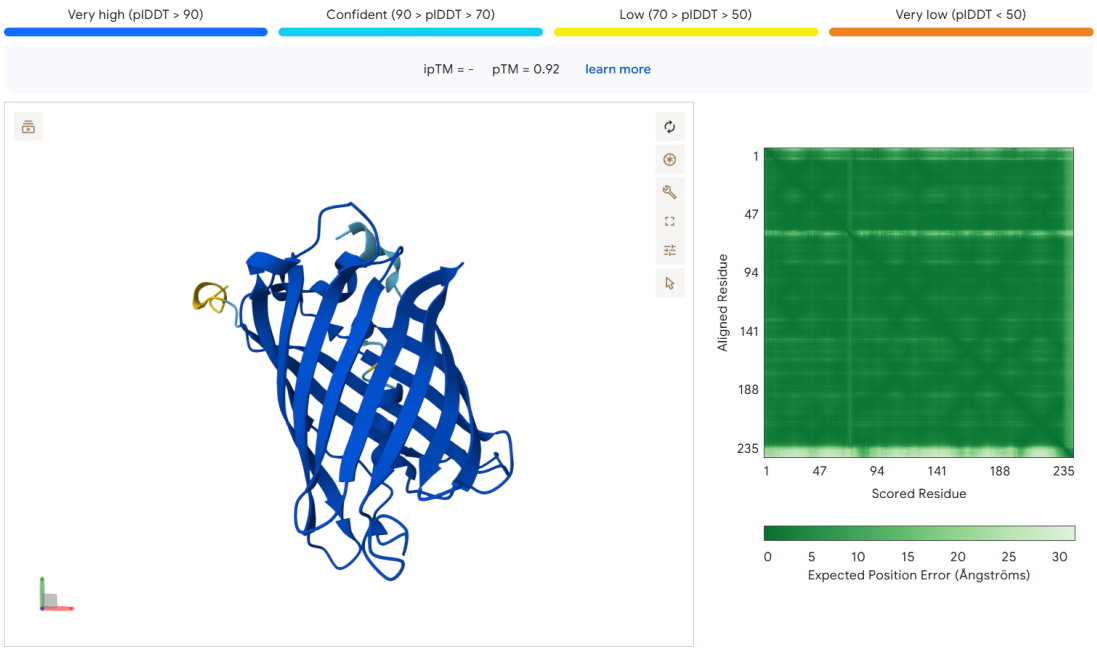
pLDDT is the predicted Local Distance Difference Test. It is a per-atom confidence score on a 0-100 scale, reported here as the mean over all atoms. It is a local measure that predicts how well each atom's immediate environment would match an experimental structure. All of these were above 90 when taking the mean, which is optimally high, and that is why the rendered structures are almost uniformly dark blue.

pTM is the predicted Template Modeling score. This estimates TM-score between the prediction and the true structure on a 0-1 scale. Unlike pLDDT, this is a global measure that asks the question of whether the general fold is probably correct. Generally, above 0.8 indicates high confidence in the whole topology. Here it would showcase that the complete beta-barrel is coherent, not just the local geometry.

Finally, the PAE, or Predicted Aligned Error, is the panel beside each structure that gives expected positional error in ångströms. For every residue pair. Uniformly low (or green) PAE means the model is confident about how the whole chain packs together. This is seen below in the AlphaFold Output.

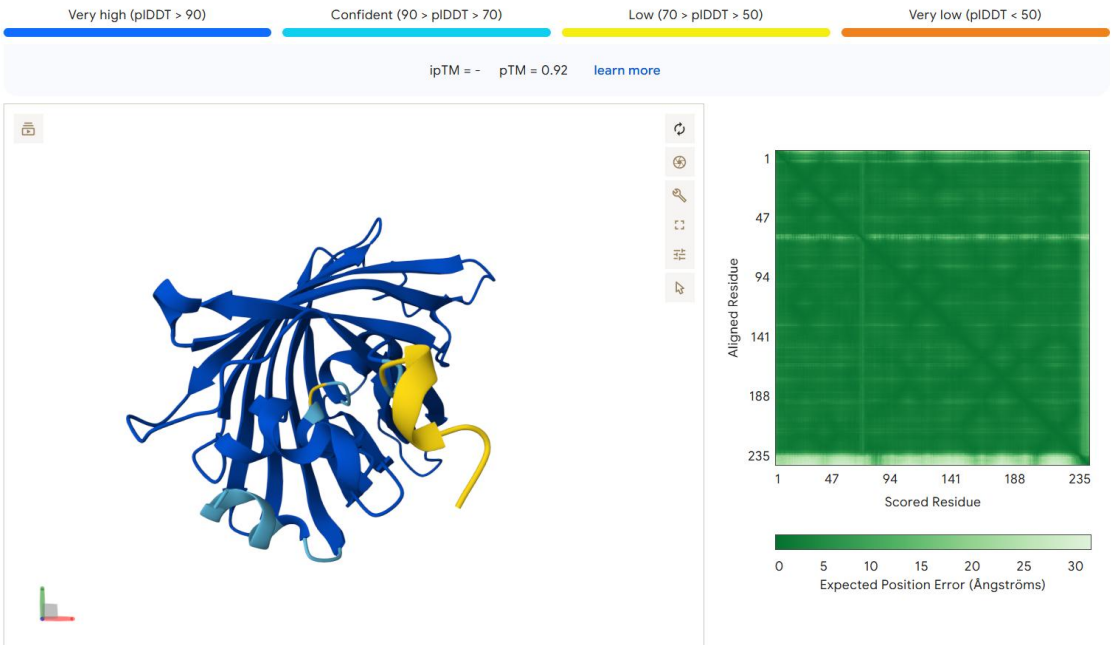
AlphaFold Output:

Design_022:



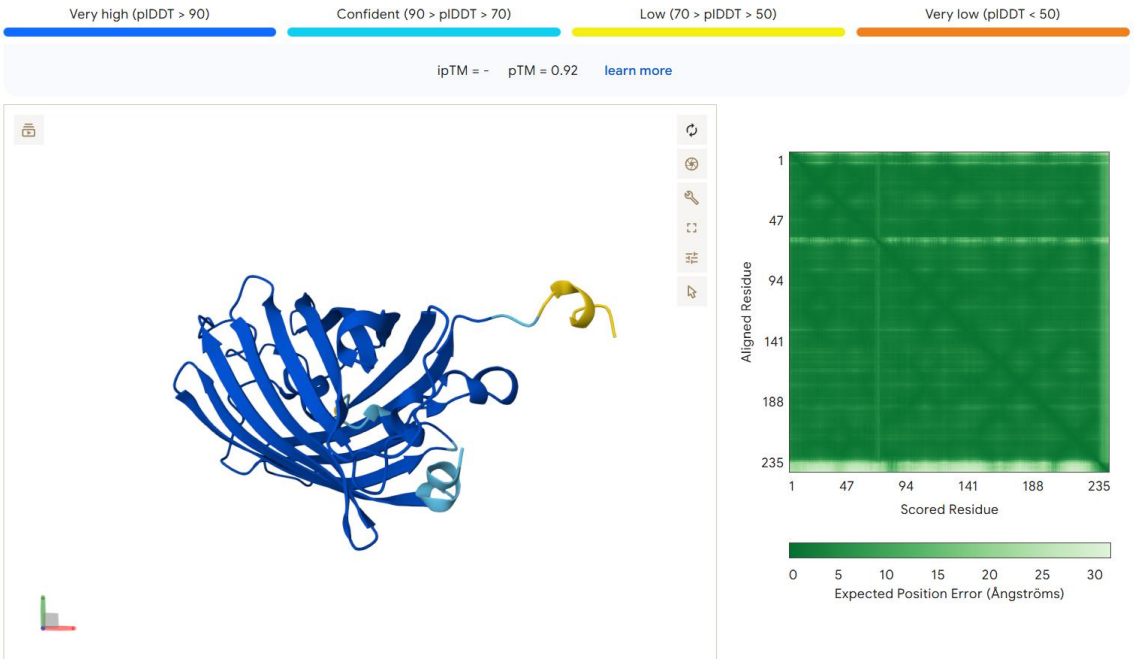
SKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDVAYGKLTLKFICTTGKLPVPWPTLV
TTLAYGVQCFSRYPDHMKQHDFFKSAVPEGYVRERTIFFKDDGNYKTRAEVKFEGDTLV
NRIELKGIDFKEDGNILGHKLEYNYNSHNVYIMADKQKNGIKVNFKIRHNIEDGSVQLA
DHYQQNTPIGDGPVLLPDNHYLSTQSALSKDPNEKRDHMVLLEFVTAAGITHGMDELY
K

Design_027:



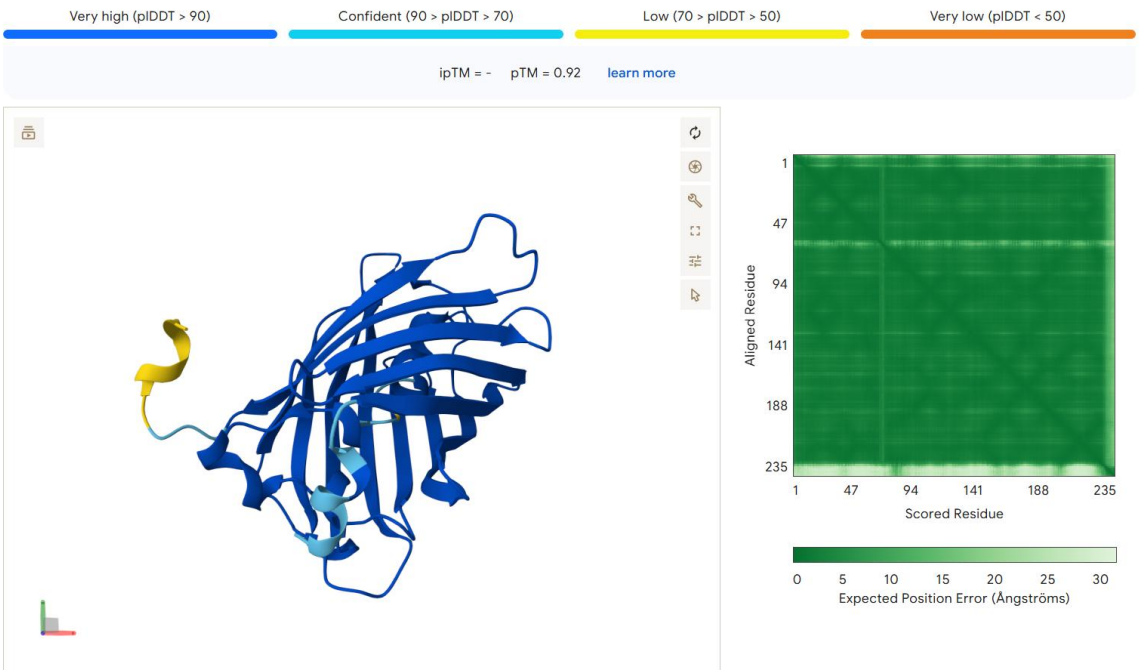
SKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDTTYGKLTLKFICTTGKLPMPWPTLV
TTLSYGVQCFSRYPDHMKQHDFFKSAVPEGYVRERTIFFKDDGNYKTRAEVKFEGDTLV
NRIELKGIDFKEDGNILGHKLEYNYNSHNVYIMADKQKNGIKVNFKIRHNIEDGSVQLA
DHYQQNTPIGDGPVLLPDNHYLSTQSALSKDPNEKRDHMVLLEFVTAAGITHGMDELY
K

Design_018:



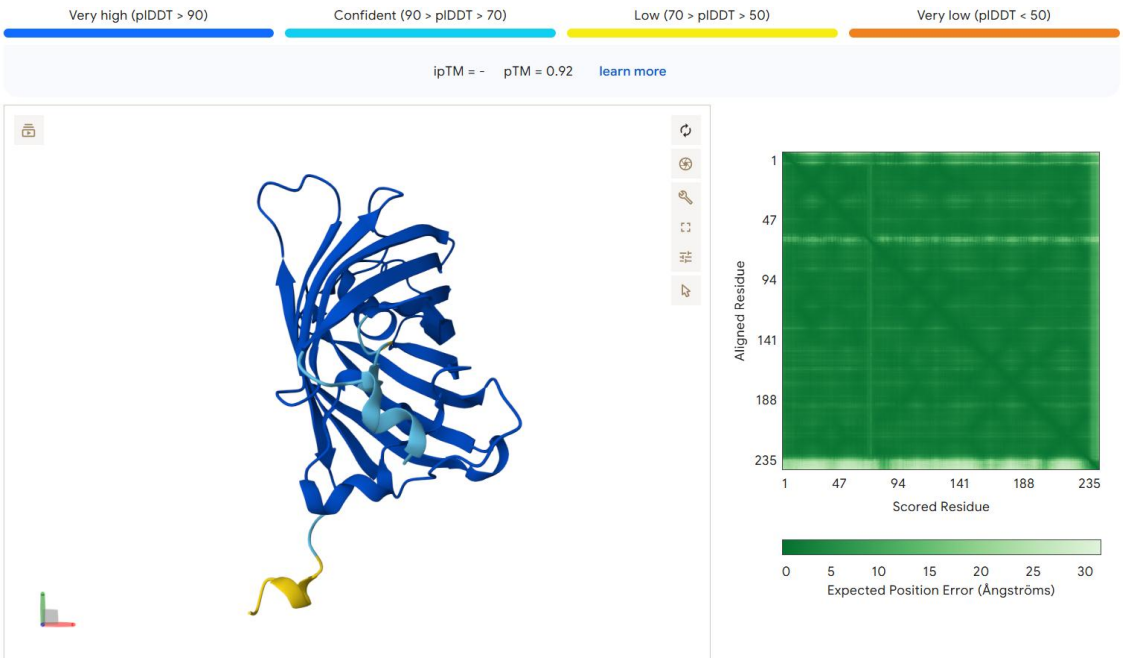
SKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDVTHGKLTLKFICTTGKLPVPWPTLV
TTLSYGVQCFSRYPDHMKQHDFFKSAVPEGYVRERTIFFKDDGNYKTRAEVKFEGDTLV
NRIELKGIDFKEDGNILGHKLEYNYNSHNVYIMADKQKNGIKVNFKIRHNIEDGSVQLA
DHYQQNTPIGDGPVLLPDNHYLSTQSALSKDPNEKRDHMVLLEFVTAAGITHGMDELY
K

Design_092:



SKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDVTYGKLTLKFICTTGKLPVPWPTLV
TTLSYGVQCFSRYPDHMKQHDFFKSAVPEGYVRERTIFFKDDGNYKTRAEVKFEGDTLV
NRIELKGIDFKEDGNILGHKLEYNYNSHNVYIMADKQKNGIKVNFKIRHNIEDGSVQLA
DHYQQNTPIGDGPVLLPDNHYLSTQSALSKDPNEKRDHMVLLEFVTAAGITHGMDELY
K

Design_061:



SKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDDTNGKLTLKFICTTGKLPVPWPTLV
TALSYGVQCFSRYPDHMKQHDFFKSAVPEGYVRERTIFFKDDGNYKTRAEVKFEGDTLV
NRIELKGIDFKEDGNILGHKLEYNYNSHNVYIMADKQKNGIKVNFKIRHNIEDGSVQLA
DHYQQNTPIGDGPVLLPDNHYLSTQSALSKDPNEKRDHMVLLEFVTAAGITHGMDELY
K

Divergence Analysis

Most frequent residue at each position over all 47,854 training sequences is the chromophore-forming triad SYG highlighted below in the consensus sequence:
SKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDATYGKLTLKFICTTGKLPVPWPTLV
TTL**SYG**VQCFSRYPDHMKQHDFFKSAVPEGYVRERTIFFKDDGNYKTRAEVKFEGDTLV
NRIELKGIDFKEDGNILGHKLEYNYNSHNVYIMADKQKNGIKVNFKIRHNIEDGSVQLA
DHYQQNTPIGDGPVLLPDNHYLSTQSALSKDPNEKRDHMVLLEFVTAAGITHGMDELY
K

Substitution Table:

Every position at which a top-5 candidate differs from the consensus. Conservation is how often the consensus residue occurs at that position in the training set. There were 11 total substitutions across the top 5 and 11 were inherited from the prompt and 0 were generated by the model.

Rank	Design	Substitution	Position	Conservation	Source
1	Design_022	A36V	36	0.999	prompt
1	Design_022	T37A	37	0.980	prompt
1	Design_022	S64A	64	0.986	prompt
2	Design_027	A36T	36	0.999	prompt
2	Design_027	V54M	54	0.987	prompt
3	Design_018	A36V	36	0.999	prompt
3	Design_018	Y38H	38	0.980	prompt
4	Design_092	A36V	36	0.999	prompt
5	Design_061	A36D	36	0.999	prompt
5	Design_061	Y38N	38	0.980	prompt
5	Design_061	T62A	62	0.986	prompt

Positional Distribution:

Where do the substitutions fall across the 237 residues. The prompt covers positions 1-64, and everything from 65 onward is the model output. As looked upon above all substitutions fall within the prompt region.

Design	1-39	40-79	80-118	119-158	159-197	198-237
Design_022	2	1	0	0	0	0
Design_027	1	1	0	0	0	0
Design_018	2	0	0	0	0	0
Design_092	1	0	0	0	0	0
Design_061	2	1	0	0	0	0
Total	8	3	0	0	0	0

Decoding Strategy:

I used the top-p sampling strategy. The p was set to 0.8. It was then implemented in the inference.py. At each position the model produces a probability over the 20 amino acids. Top-p sampling would sort those descending, accumulates probability mass until it exceeds p = 0.8, discards every residue beyond that point, and samples from the ones that remain. The alternative that I also worked with was greedy decoding that takes the highest probability residue every time and is therefore fully deterministic. Here, the same prompt would always yield the same protein.

I decided to choose top-p because it allows the model to contribute something of its own, though the effects were still relatively small here. However, checking greedy it had 0 substitutions beyond the prompt for all designs while top-p was able to have 5 beyond the prompt.

For the other remaining parameters that were used, max sequence length was set to 512 tokens. Also, the batch size was one in this inference as well. Finally, the seed was set to 0.

Interpretation:

Finetuning was able to succeed at this task. Every candidate folds into a confident GFP beta-barrel (pTM 0.92, mean pLDDT above 90), terminates at exactly 237 residues, and generally preserves the SYG chromophore triad. This includes the tyrosine at position 65 which would be in the model generated output. However, it would seem that the model generally just contributed to the consensus sequence and didn't add anything of its own. All 11 substitutions in the top 5 were handed to it in the prompt. Across all 92 designs, 188 prompt region substitutions were made with only 5 from the decoder.

This seems to follow the training data, which is a point-mutation scan of a single protein at 98.4% mean per position conservation: there is almost no entropy left for a converged autoregressive model to sample, so the next-residue distribution is sharp enough that even nucleus sampling usually finds one residue occupying the entire probability mass.

For the pLDDT ranking, it seemed to distinguish very little. The top 5 span 0.26 pLDDT points while differing from one another by only 1 to 3 residues out of 237. This gap is small

comparable to AlphaFold's own run-to-run variation, so the ordering is better read as noise than as a quality gradient. The pLDDT did work as a pass/fail filter to confirm the sequences are foldable, which is what the pTM > 0.8 gate was generally for, and not as a way to rank somewhat near identical candidates against each other.